NC STATE UNIVERSITY

AI Student Symposium 2026

ECE 492/525 Poster NNDL.57
Project ID #36

Comparative Study of CNN and AE Models for Pneumonia Detection

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Problem Statement:

Neural network detection of pneumonia to save dangerous missed diagnosis

Data:

Chest X-Ray Images (Pneumonia) Dataset from Kaggle, which contains labeled grayscale images categorized as Normal or Pneumonia.

Approach:

- **CNN:** supervised model for direct classification
- **Autoencoder:** trained on normal images; detects pneumonia using reconstruction error
- **Goal:** compare classification vs anomaly detection

Network Architecture:

Convolutional Neural Network (CNN) vs Autoencoder (AE)

Metrics

- Accuracy, Precision, Recall, F1-score, Confusion Matrix

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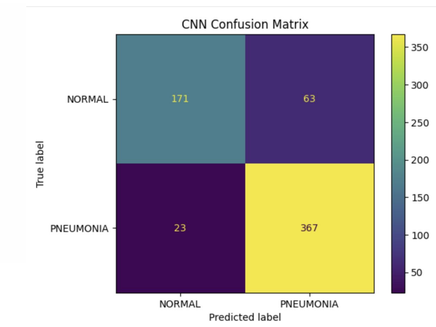
CNN Test Results
Accuracy  : 0.8622
Precision : 0.8535
Recall    : 0.9410
F1-Score  : 0.8951
```

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— Autoencoder Test
  Accuracy : 0.5817
  Precision: 0.6201
  Recall   : 0.8538
  F1-Score : 0.7184
```

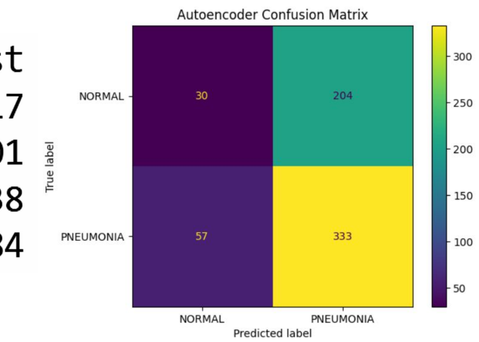

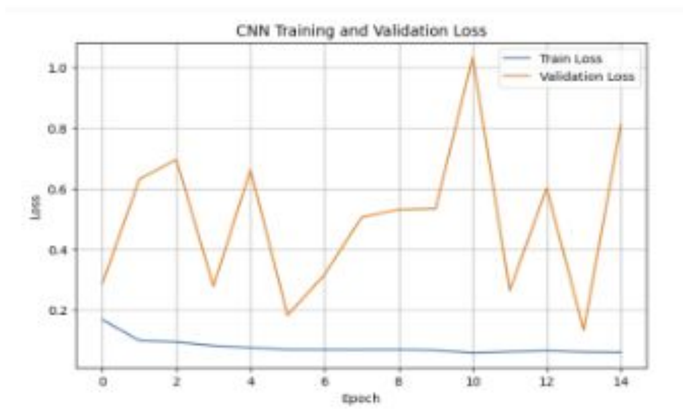

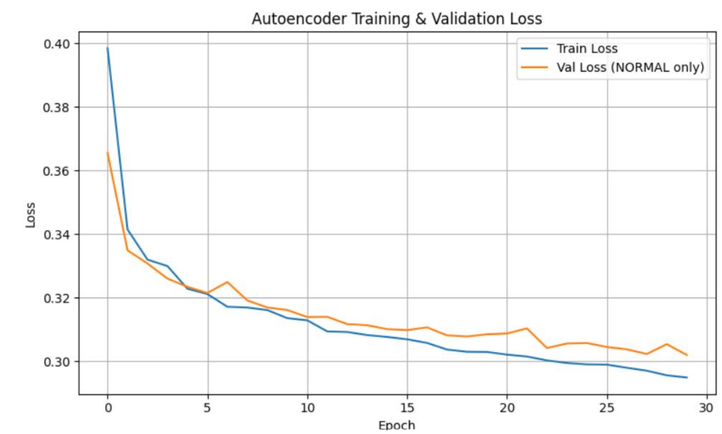

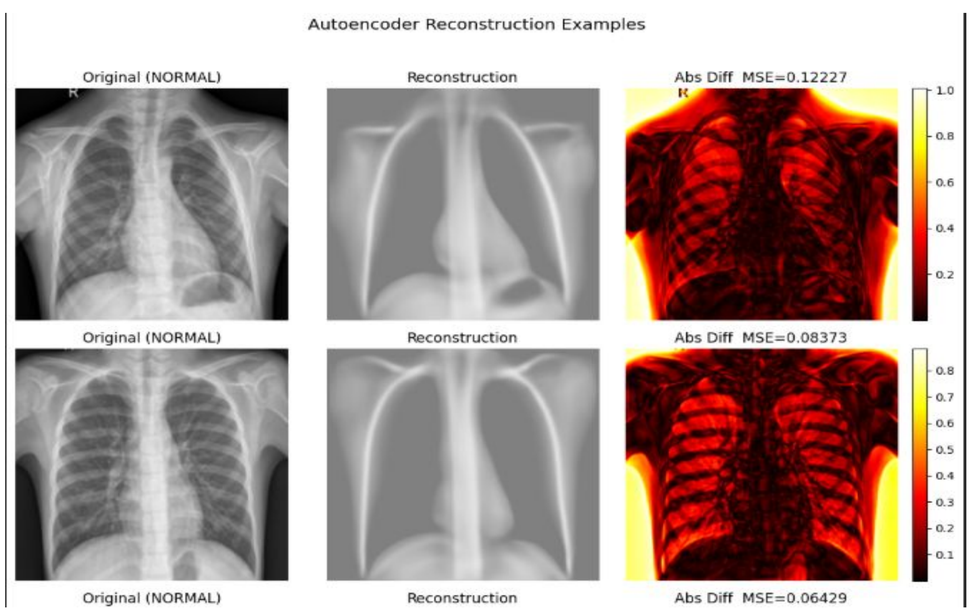

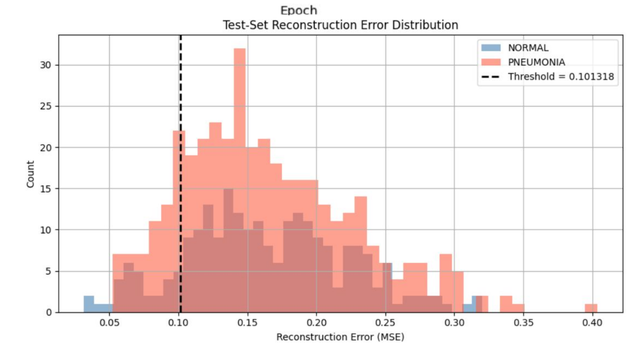

Main References:

[1] Chest X-Ray Images (Pneumonia) Dataset, https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia/data

[2] Current Diagnostic Techniques for Pneumonia, https://pmc.ncbi.nlm.nih.gov/articles/PMC11244398/